A valid node of the SEM forcing yields an extension of the underlying condition

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Editorial note. The counted unit is the article's own Lemma lem:sem-wf-valid-leaf-force-positive at source lines 841-844 together with the proof environment that begins on the next line (845-850); the pairing is the ordinary adjacent one.

Editorial note. The retained context at source lines 819-829 is the author's own definition of a node of a condition, of the union pi of a node, of validity and of satisfaction of a Sigma^0_1 formula; the retained context at source lines 831-833 is the statement of the immediately preceding transitivity lemma, which the counted proof invokes and which is counted by a sibling card of this lane.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, his natural-operation macros at source lines 104-107 and his quotation macro at source line 116, and it retains mathabx from source line 103 and cleveref, which the assembler only adds by itself when it generates navigation stubs. The LS2 font-encoding block at source lines 93-99 is not retained because the host TeX Live has no stix directory, and the inputenc load with the utf8x option at source line 8 is not retained because it fights the forced xelatex engine.

Editorial note. No retained line contains a citation, so the material carries no bibliography; the only cross-reference of the retained proof is the author Cref call to the transitivity lemma, whose statement is retained in the context.

Editorial note. The article's conservation theorems, its COH and WKL lemmas and the other lemmas of the SEM section are not counted here and are not used as premises.

Allowed standard prerequisites (editorial).

- Node of a condition, validity and the set pi of a node. A (sigma, a)-node is an M-finite sequence of pairs (i_s, rho_s) with each rho_s a non-empty M-finite R-transitive subset of X_a, increasing in the order of the integers and comparable in the direction prescribed by the bit i_s; it is valid when every block is contained in the corresponding side A_{i_s}(R) of the colouring; pi of the node is the union of sigma with all blocks; the author's

own definition is retained at source lines 819-829.. Fixes the objects of the counted lemma, namely a valid node and its union pi, whose transitivity is what makes the pair (pi, b) a condition.

- Stability of the colouring R and the collection scheme. Stability of R means that for every x in X_a the set of y above x on which R holds is either finite or cofinite, which turns the validity of a node into infinitary comparability statements; the collection scheme for Sigma^0_2 formulas then lets the finitely many bounds obtained that way be taken uniformly.. The proof converts validity into the statements that each block is below or above almost all of X_a, and then uses collection to find a single initial segment whose removal makes the comparisons uniform.
- The set attached to a node is transitive. For every node the union of sigma with all its blocks is R-transitive; this is the statement of the immediately preceding lemma, retained with this card at source lines 831-833.. The proof invokes it to conclude that the set tau it constructs is R-transitive, which is one of the two conditions for (tau, b) to be an extension of (sigma, a).

Definition 0.1. *Fix a condition* (σ, a).

(1) *A* (σ, a)-node *is an* M*-finite sequence of the form*

$$\nu = \langle (i_0, \rho_0), (i_1, \rho_1), \dots (i_{q-1}, \rho_{q-1}) \rangle$$

such that for every $s < q$, $i_s < 2$, $\rho_s \subseteq X_a$ *is a non-empty* M*-finite* R*-transitive set with* $\min \rho_s > \max \rho_{s-1}$ *(letting* $\rho_{-1} = \sigma$*), and for every* $s < t < q$, *if* $i_s = 0$ *then* $\rho_s \to_R \rho_t$, *and if* $i_s = 1$ *then* $\rho_t \to_R \rho_s$. *We then write*

$$\pi(\nu) = \sigma \cup \rho_0 \cup \dots \cup \rho_{q-1}$$

(2) *This* (σ, a)*-node is* valid *if for every* $s < q$, $\rho_s \subseteq A_{i_s}(R)$

(3) *A* (σ, a)*-node* ν satisfies *a* $\Sigma^0_1(\mathcal{M})$*-formula* $\varphi(G, x) \equiv \exists i \psi(G\restriction i, x)$ *if* $\psi(\pi(\nu), |\nu| - 1)$ *holds.*

Lemma 0.2. *Fix a* (σ, a)*-node* ν. *Then* $\pi(\nu)$ *is* R*-transitive.*

Lemma 0.3. *Fix a valid* (σ, a)*-node* ν. *Then* $(\pi(\nu), b)$ *is an extension of* (σ, a) *for some* $b \in M$.

Proof. Say $\nu = \langle (i_0, \rho_0), (i_1, \rho_1), \dots (i_{q-1}, \rho_{q-1}) \rangle$. By definition of validity and stability of R, for every $s < q$, if $i_s = 0$ then $\forall x \in \rho_s \forall^\infty y \in X_a$, $R(x, y)$, and if $i_s = 1$ then $\forall x \in \rho_s \forall^\infty y \in X_a$, $R(y, x)$. By $\mathsf{B}\Sigma^0_2$, there exists some $r \in M$ such that for every $s < q$, if $i_s = 0$ then $\rho_s \to_R X_a \backslash \{0, \dots, r\}$, and if $i_\ell = 1$ then $X_a \backslash \{0, \dots, r\} \to_R \rho_s$. Let $b \in M$ be such that $X_b = X_a \backslash \{0, \dots, r\}$ and let $\tau = \sigma \cup \rho_0 \cup \dots \cup \rho_{t-1}$. By Lemma 0.2, τ is R-transitive and by construction, $X_b \subseteq X_a$ is included in a minimal interval of τ. Thus (τ, b) is an extension of (σ, a). □